

Lecture #11

Content to be covered

- ❑ Polarization
 - ❖ What is polarization?
 - ❖ Plane of Vibration and Plane of Polarization
- ❑ Production of polarized light
 - ❖ By reflection
 - ❖ By refraction

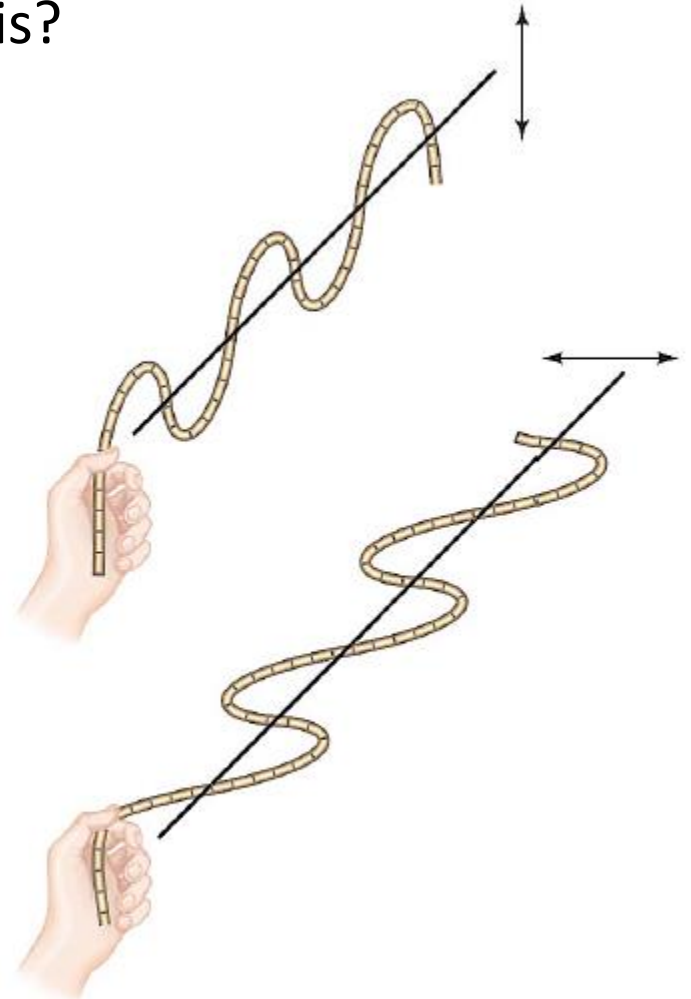
Resources to be consulted

- ❑ OPTICS by Ajay Ghatak
- ❑ <http://vlab.amrita.edu/?sub=1&brch=189&sim=333&cnt=4>
- ❑ <https://www.olympus-lifescience.com/en/microscope-resource/primer/java/polarizedlight/3dpolarized/>

Why Polarization ???

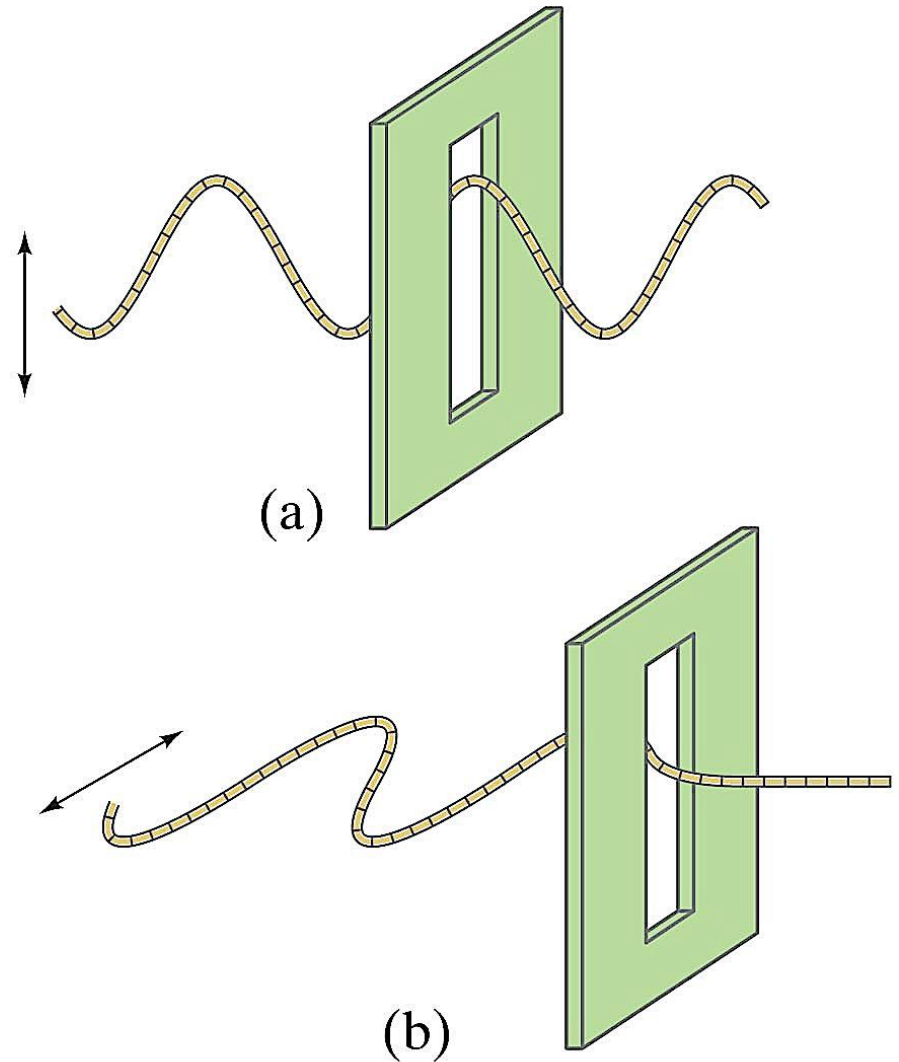
After the study of **interference and diffraction**, we know that light behaves as wave. So light is a form of wave motion. But a question still remains that what type of wave is this?

Longitudinal?
Transverse?



Polarization

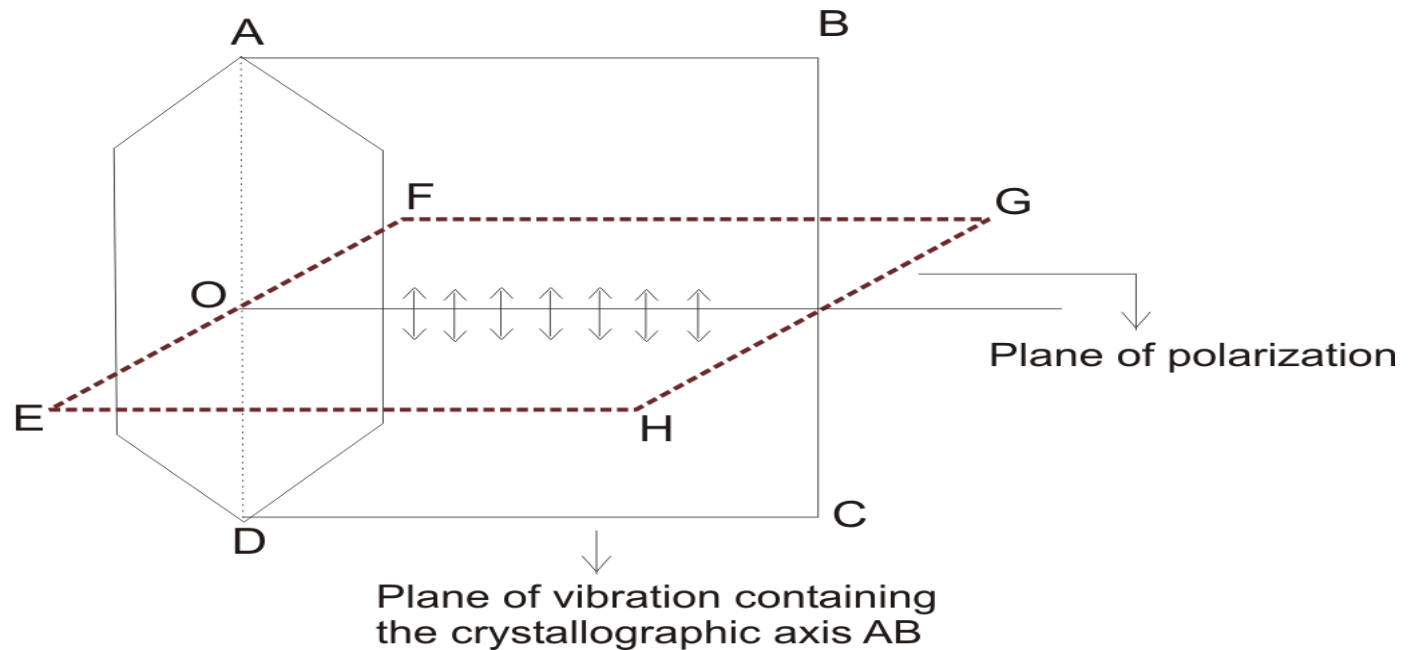
Polarized light will not be transmitted through a polarized film whose axis is perpendicular to the polarization direction.



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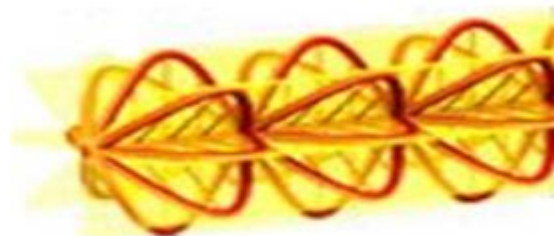
Plane of Vibration and Plane of Polarization

- The plane containing the direction of propagation of light but no vibration is called **plane of polarization**.
- The plane containing the direction of propagation of light as well as vibrations called **plane of vibration**.
- Both, plane of polarization and plane of vibration are **perpendicular to each other**.

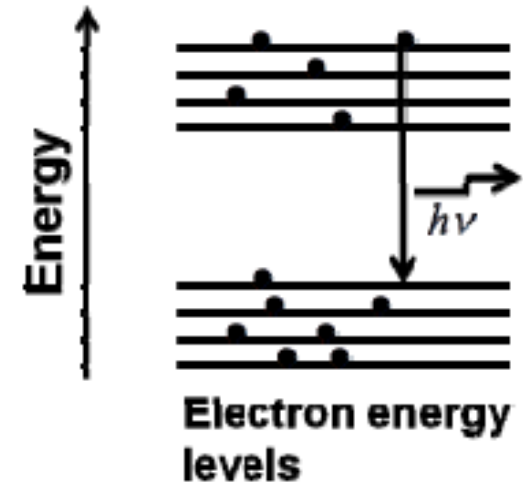


Polarization

If the plane of vibration is changed in a random manner in very short interval of time, then such a wave is known as un-polarized wave. Natural light or ordinary light is un-polarized in nature.

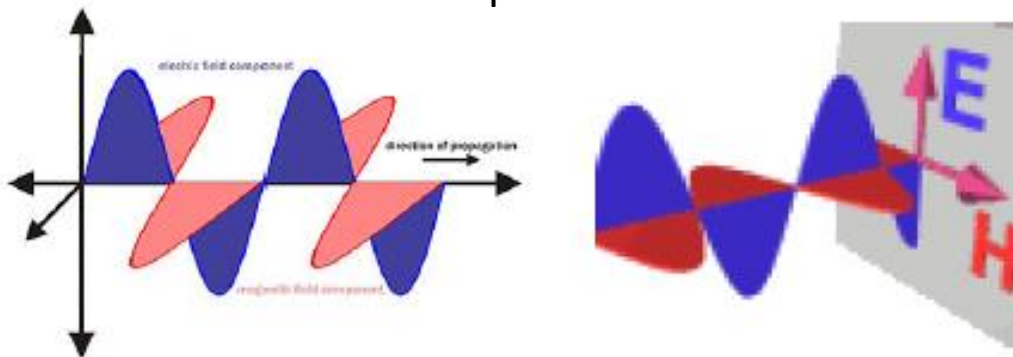


Un-polarized light



Electron energy levels

Plane polarized wave

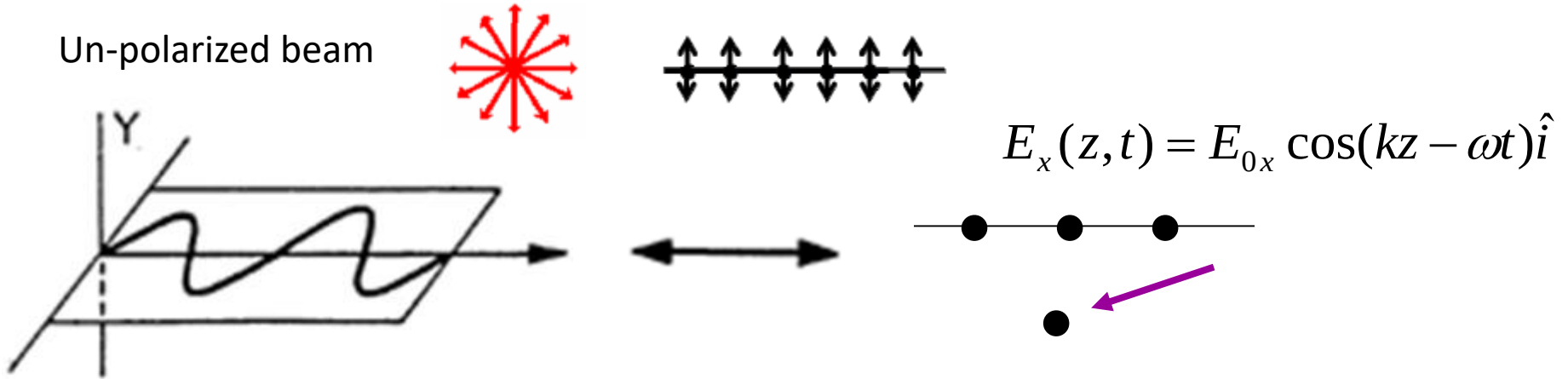


Light is polarized when its electric fields oscillate in a single plane, rather than in any direction perpendicular to the direction of propagation.

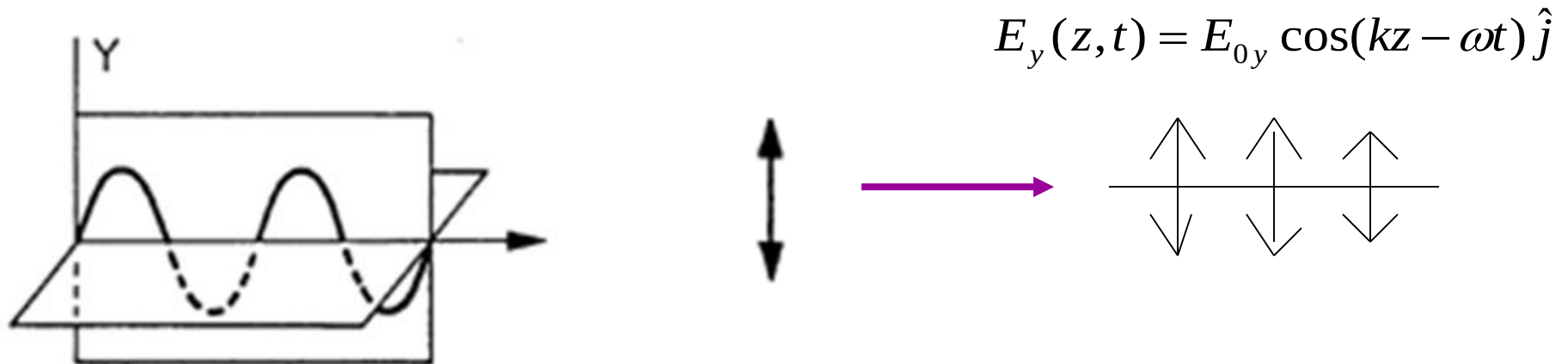
Light is an electromagnetic wave and transverse in nature.

Un-polarized & Polarized light

Un-polarized beam

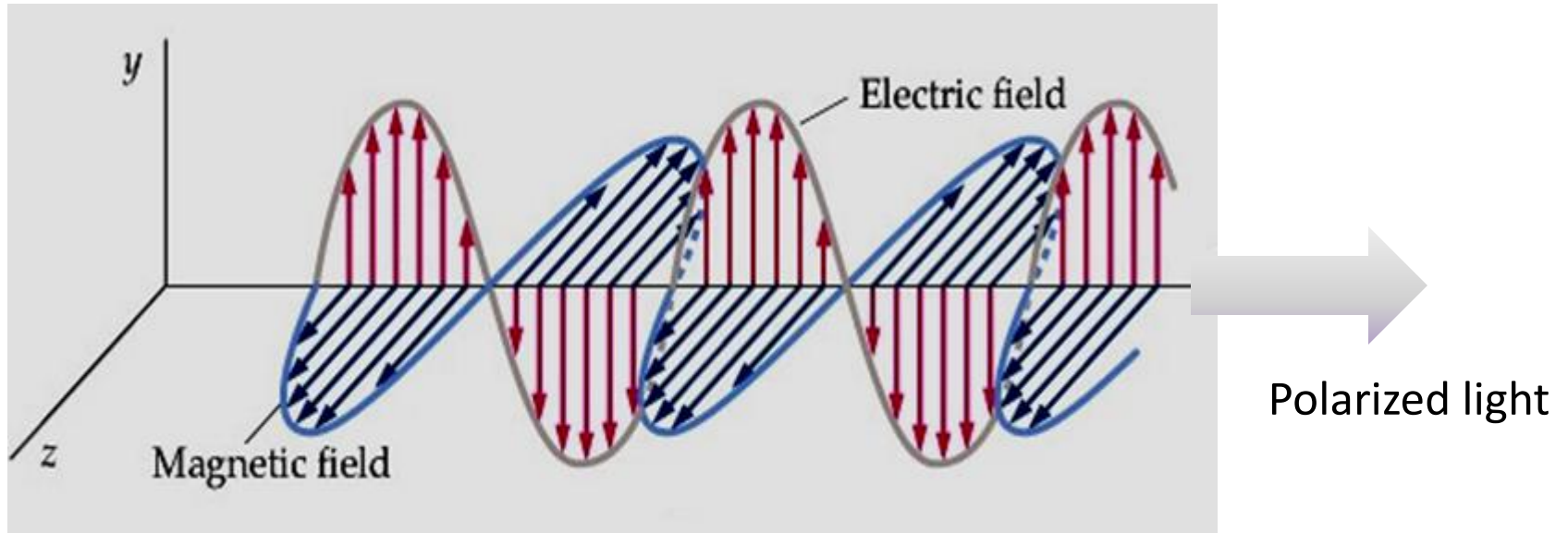


Plane polarized light with vibration **perpendicular** to the plane of paper



Plane polarized light with vibrations **parallel** to the plane of paper

Polarization



The process of transforming un-polarized light into polarized light is known as **polarization**.

Mathematical Representation of Plane polarized Light

Suppose light is propagating in z-direction,
Mathematically a plane polarized light can be represented as:

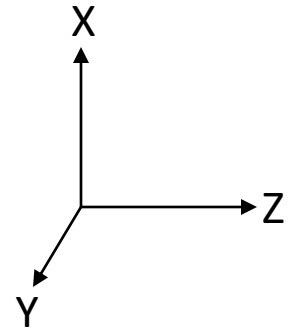
$$\vec{E}_x(z, t) = \hat{i} E_{0x} \cos(kz - \omega t)$$

or

$$\vec{E}_y(z, t) = \hat{j} E_{0y} \cos(kz - \omega t)$$

$$k = \frac{\omega}{v}$$

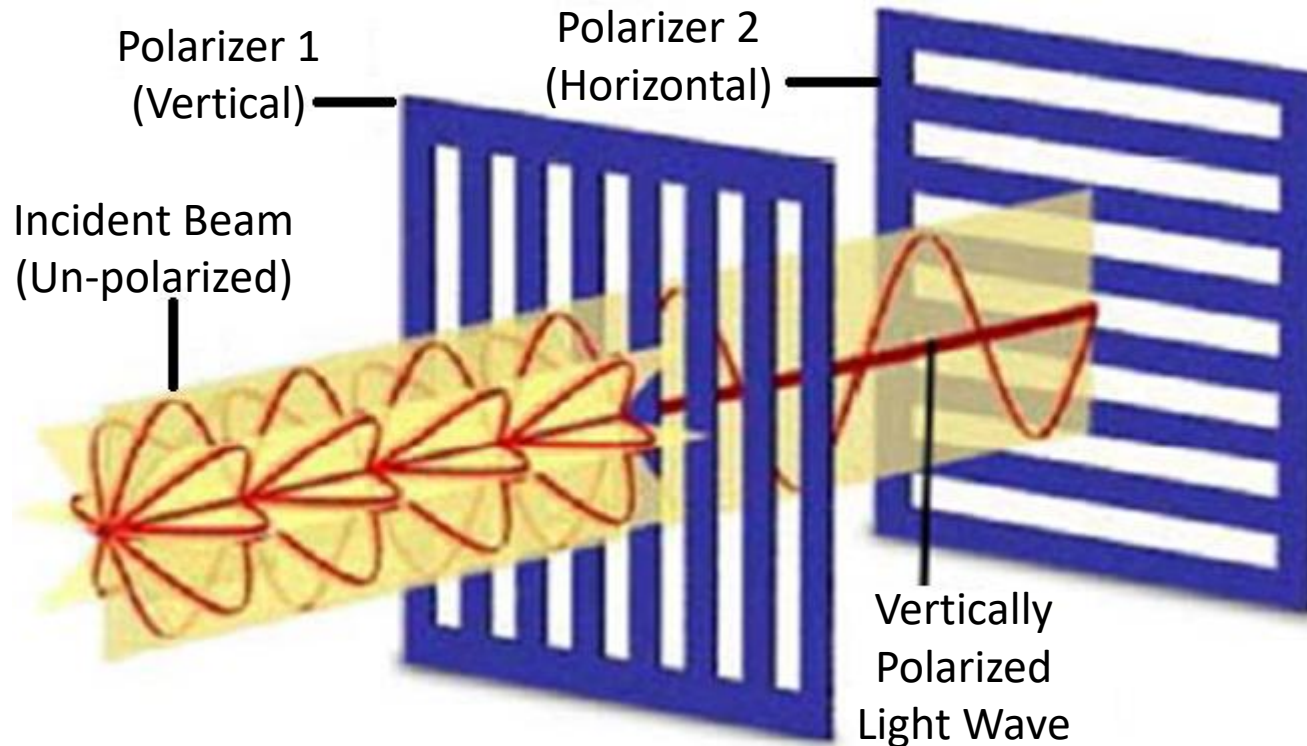
$$v = \frac{1}{\sqrt{\epsilon\mu}}$$



Polarized light consists of waves which are vibrating in one direction only.

Polarizer & Analyzer

Consider an un-polarized light wave is passing through crossed polarizers.



This means that if initially un-polarized light passes through crossed polarizer, no light will get through the second one.

Conclusion: **Light is a transverse wave**

Production of Polarized Light

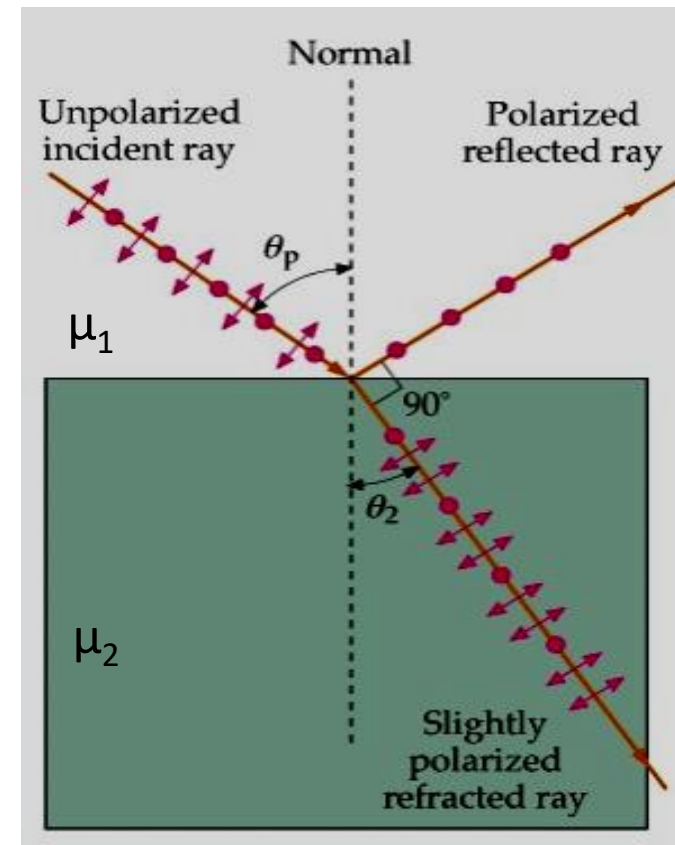
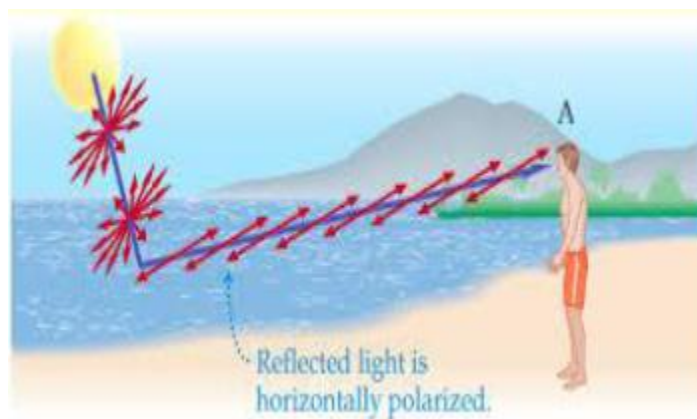
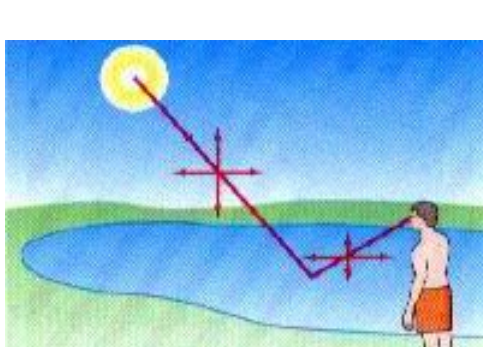
1. By Reflection: Brewster's Law
2. By Refraction: Malus' Law
3. By Selective Absorption: Dichroic material
4. By Double Refraction:
 - Nicol Prism
 - Wave Plates

Polarization by Reflection: Brewster's Law

- In 1881, Brewster on the basis of his experimental observations discovered that when **unpolarized light** is incident at polarizing angle on the dielectric medium the reflected light is **completely plane polarized having vibration perpendicular to the plane of incidence.** The polarizing angle is different for different reflecting surfaces.
- According to him the **tangent of polarizing angle (θ_p) is equal to the refractive index of the medium that is**

$$\mu = \tan \theta_p$$

$$\theta_p = \tan^{-1} \left[\frac{\mu_2}{\mu_1} \right]$$



Polarization by Reflection: Brewster's Law

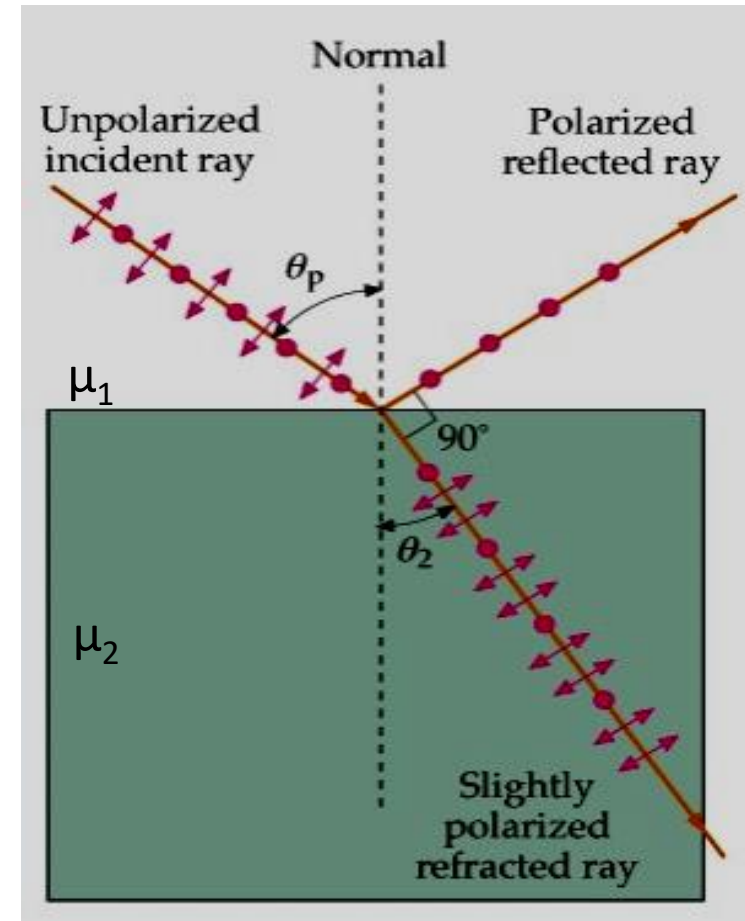
Show that $\theta_2 + \theta_p = \pi/2$

$$\tan \theta_p = \frac{\sin \theta_p}{\cos \theta_p} = \frac{\mu_2}{\mu_1} = \mu$$

As per Snell's law: $\mu_1 \sin \theta_p = \mu_2 \sin \theta_2$

Therefore $\cos \theta_p = \sin \theta_2 = \cos(90 - \theta_2)$

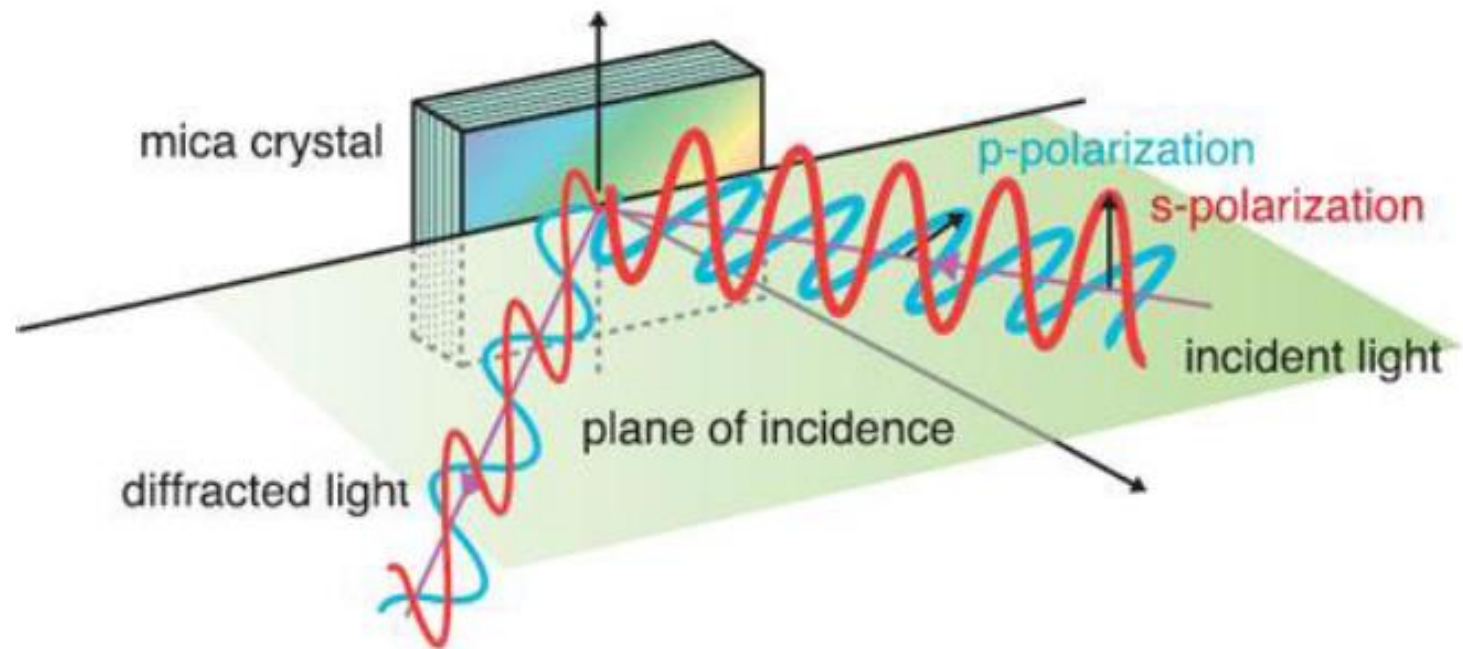
$$\Rightarrow \boxed{\theta_p + \theta_2 = \frac{\pi}{2}}$$



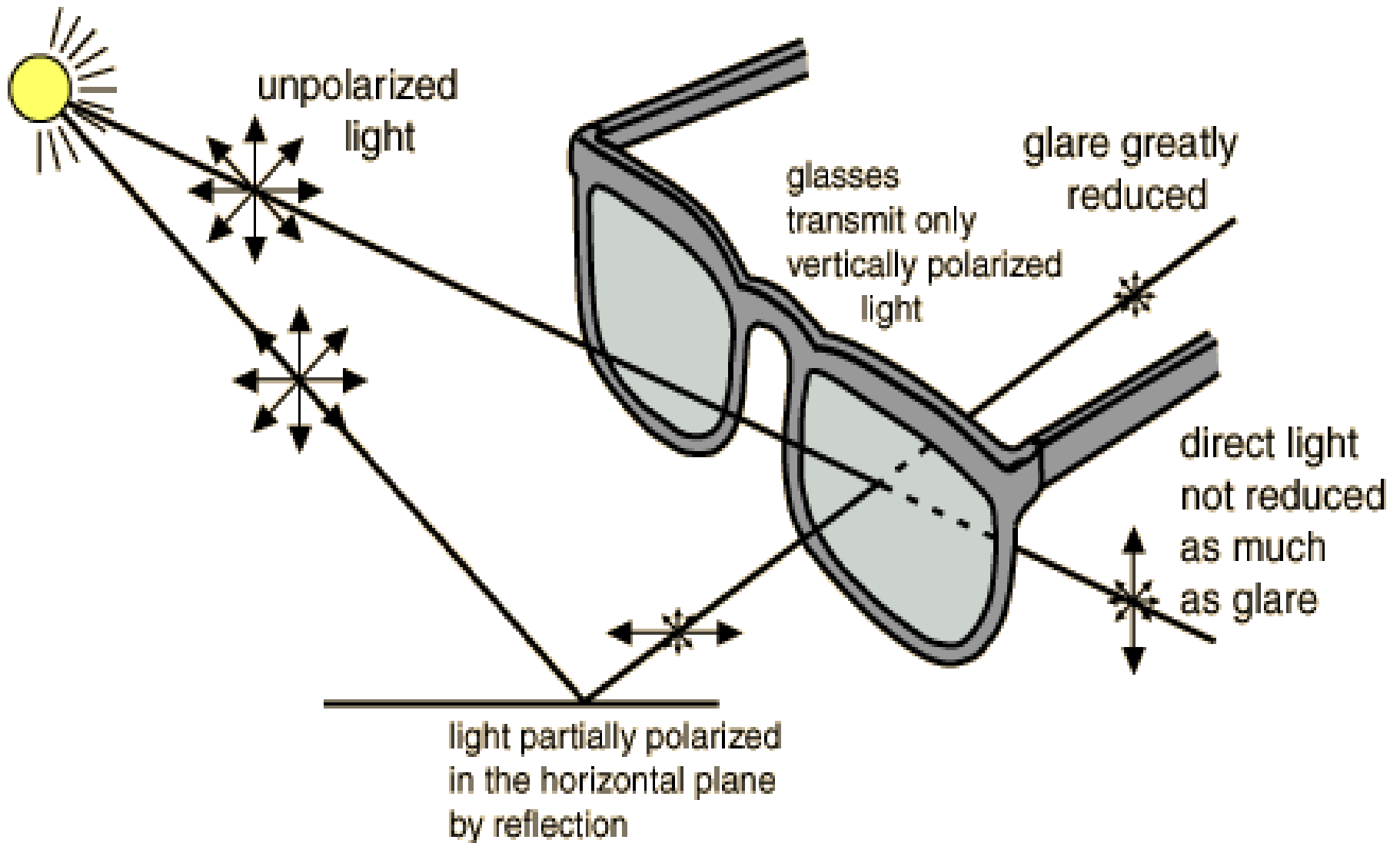
Here, reflected light has polarization vector perpendicular to the plane of incidence. Reflected and transmitted rays are at right angles to each other.

p-polarized light: parallel to plane of incidence

s-polarized light: perpendicular to plane of incidence.



Use of Polaroid

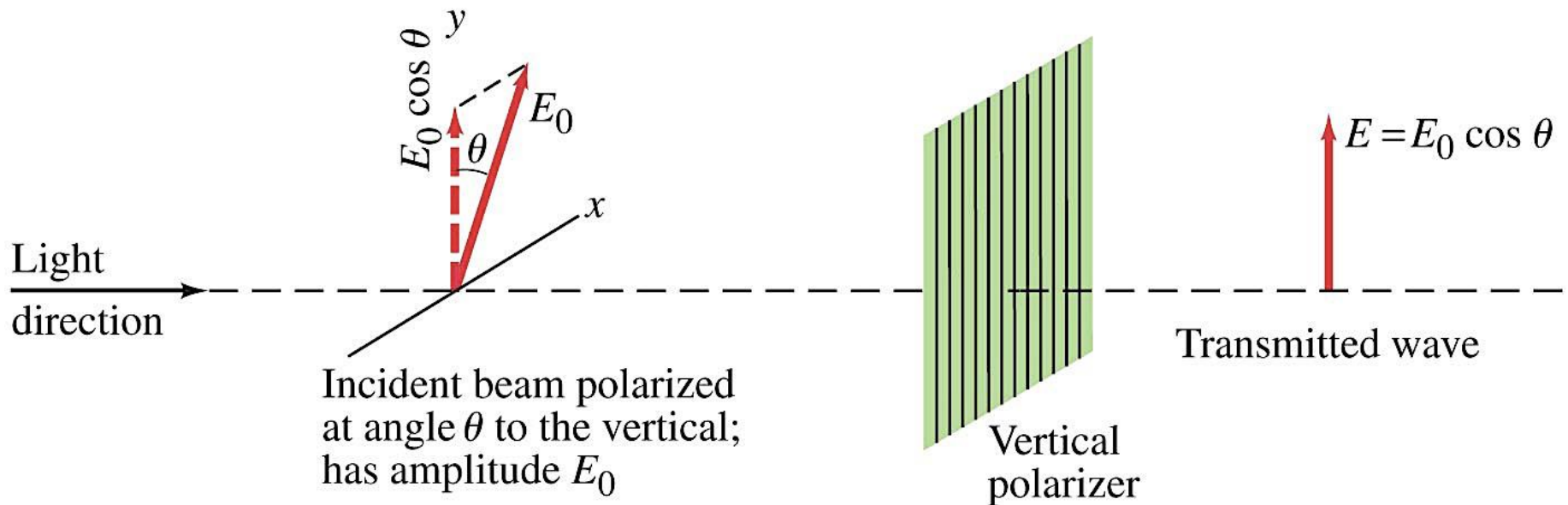


Law of Malus

When light passes through a polarizer, only the component parallel to the polarization axis is transmitted. If the incoming light is plane polarized, the outgoing intensity is:

$$I = I_0 \cos^2 \theta$$

Intensity of plane polarized wave is reduced by polarizer



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Law of Malus

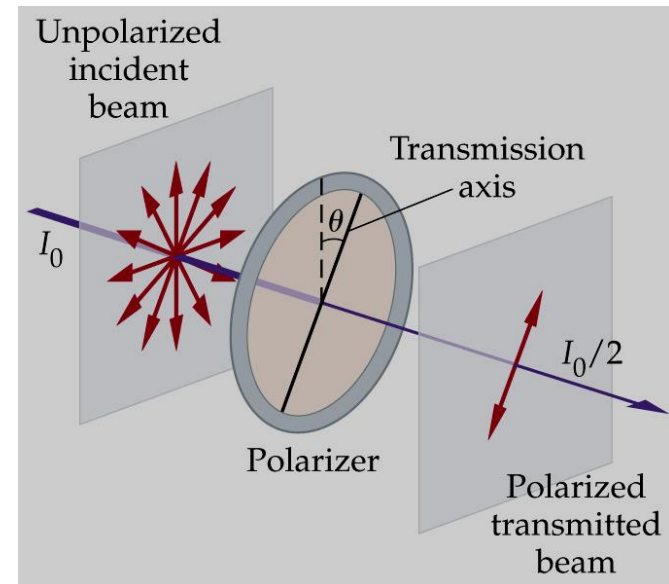
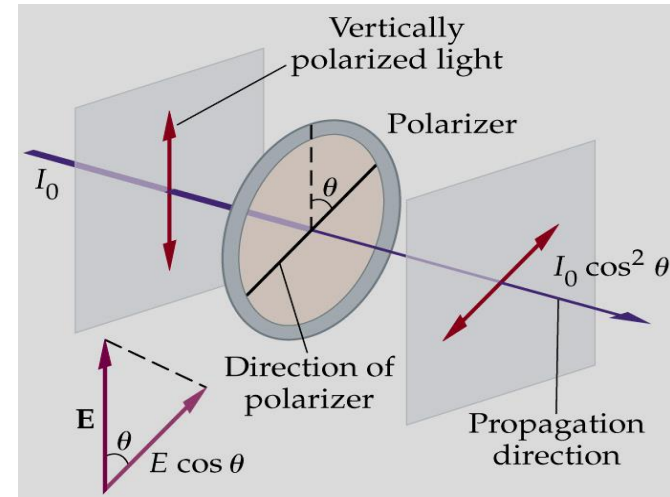
$$I = I_0 \cos^2 \theta$$

where

I is intensity of transmitted light.

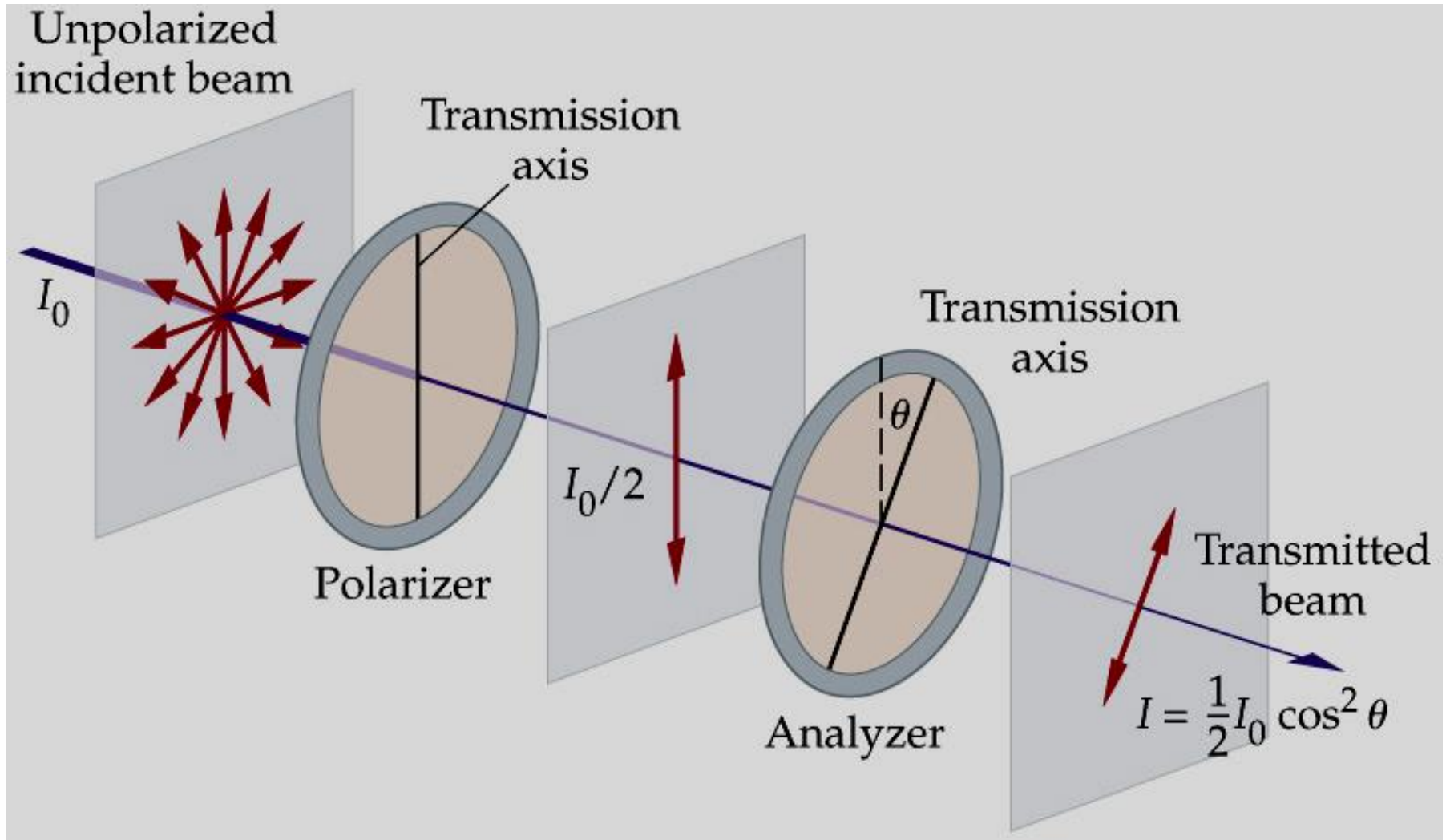
I_0 is intensity of incident light.

θ is angle between plane of incident light and direction of polarizer.



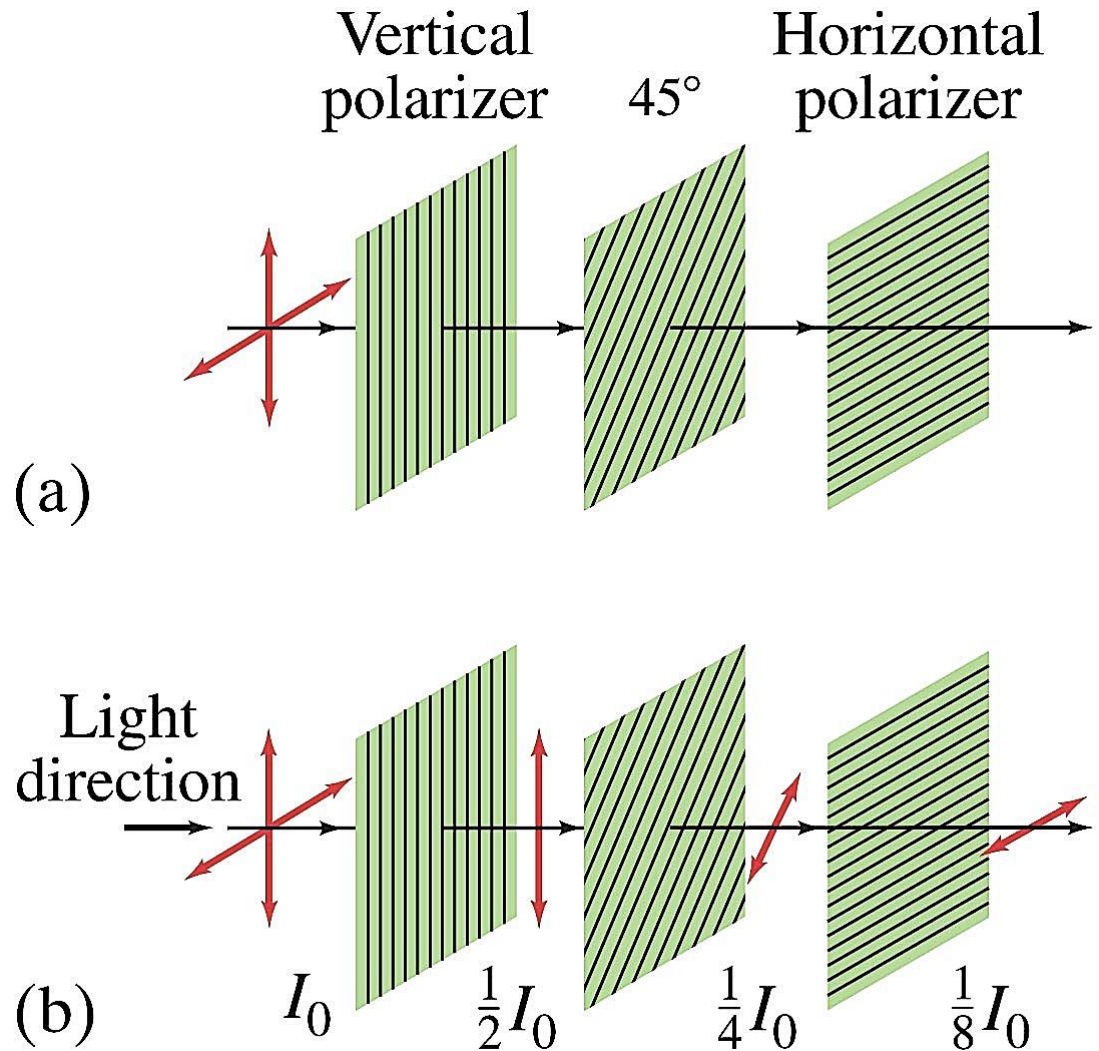
- Un-polarized light have E-field vibration in all directions.
- Therefore $I = I_0 \langle \cos^2 \theta \rangle = I_0/2$

Two Consecutive Polarizer



Three Consecutive Polarizer

When un-polarized light falls on two crossed Polaroids (axes at 90°), no light passes through. What happens if a third Polaroid, with axis at 45° to each of the other two, is placed between them?



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Problem

An un-polarized light passes through a vertically placed polarizer having horizontal polarization axis. Subsequently it passes through a polarizer with its pass axis at 90° with respect to vertical and two polarizers having their polarization axes at an angle 30° and 60° with vertical, respectively. What will be the intensity of the emergent light?

An un-polarized light passes through a vertically placed polarizer having horizontal polarization axis. Subsequently, it passes through a polarizer with its pass axis at 90° with respect to vertical followed by two polarizers having their polarization axes at an angle 30° and 0° with vertical, respectively. Calculate the intensity of the emergent light. Also, discuss the case when the last two polarizers are interchanged.